Fog and Edge Computing Presentation

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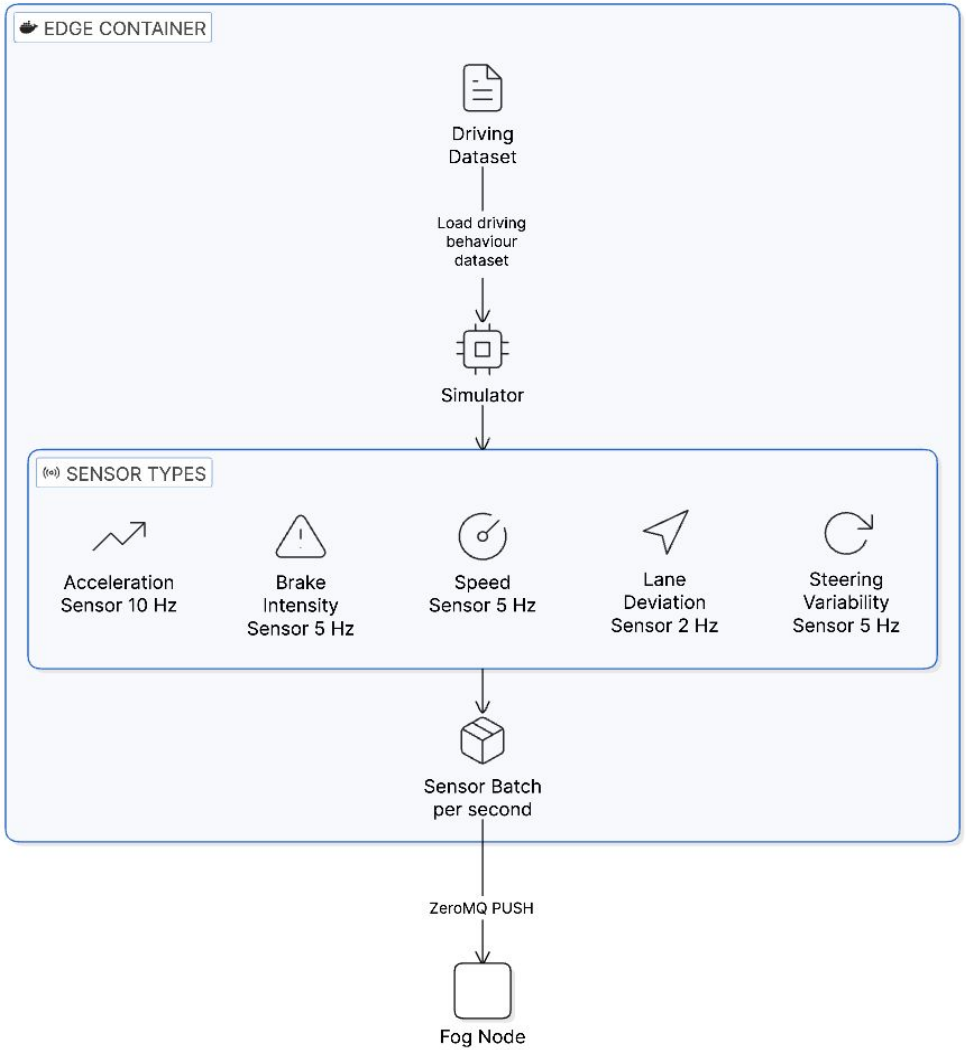
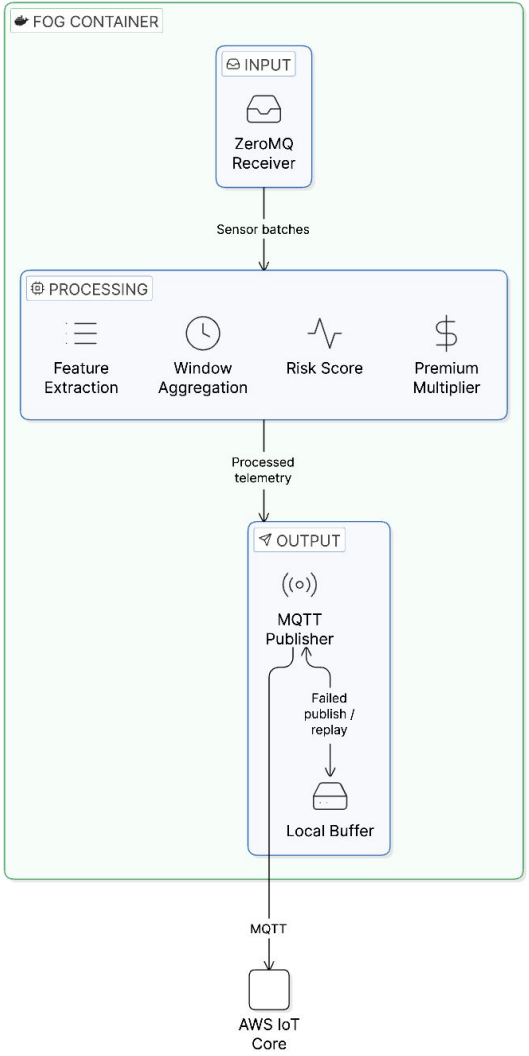
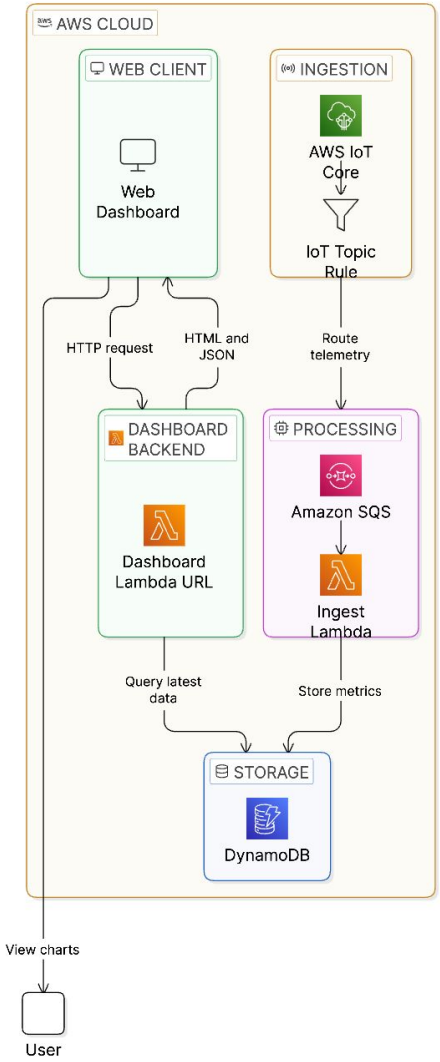
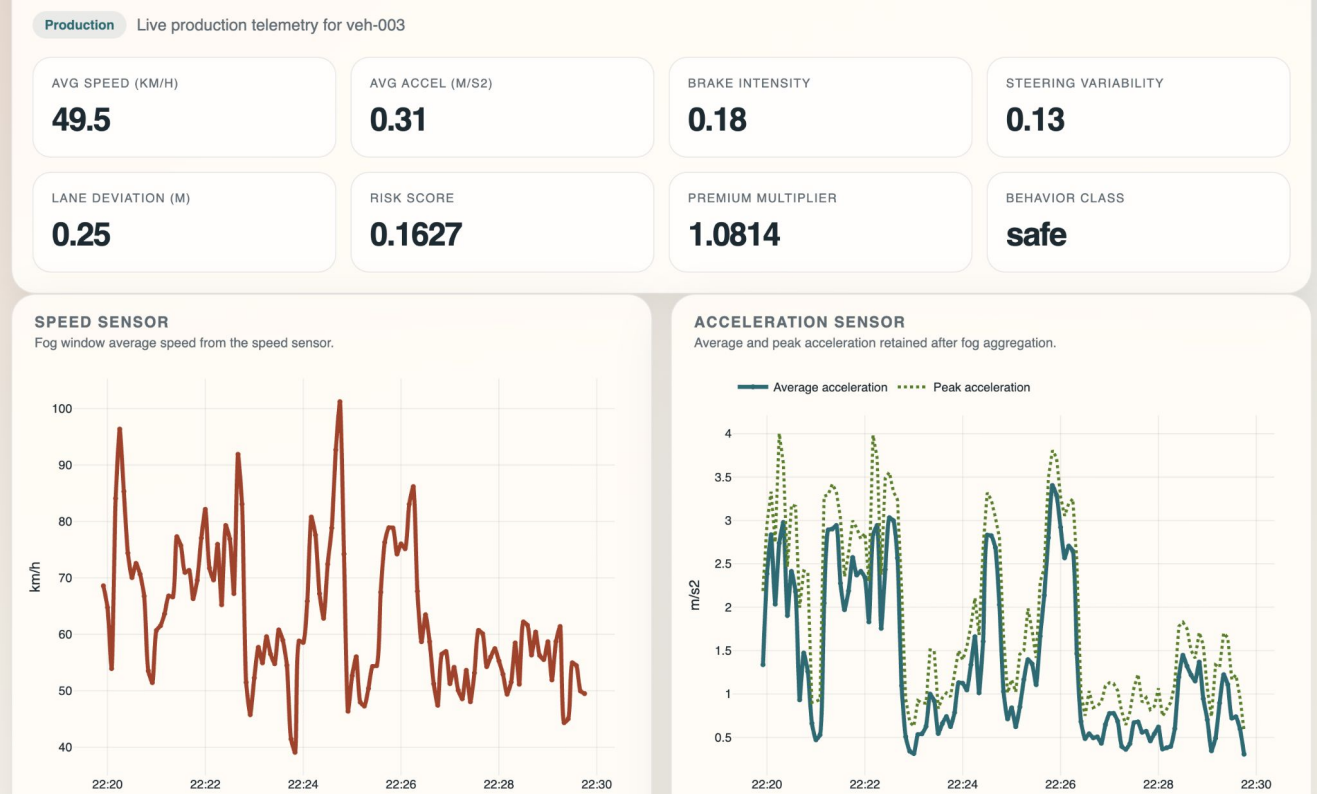
Motivation and High-level Description of the Project

- Vehicle driving behaviour
- Sensors for capture driving features
- Can be used for car insurance real-time pricing based on the driving risk evaluated from the sensor data


EDGE CONTAINER
Driving Dataset
Load driving behaviour dataset
Simulator
SENSOR TYPES
Acceleration Sensor 10 Hz
Brake Intensity Sensor 5 Hz
Speed Sensor 5 Hz
Lane Deviation Sensor 2 Hz
Steering Variability Sensor 5 Hz
Sensor Batch per second
ZeroMQ PUSH
Fog Node


FOG CONTAINER
INPUT
ZeroMQ Receiver
Sensor batches
PROCESSING
Feature Extraction
Window Aggregation
Risk Score
Premium Multiplier
Processed telemetry
OUTPUT
MQTT Publisher
Failed publish / replay
Local Buffer
MQTT
AWS IoT Core

Application Architecture



Application Overview